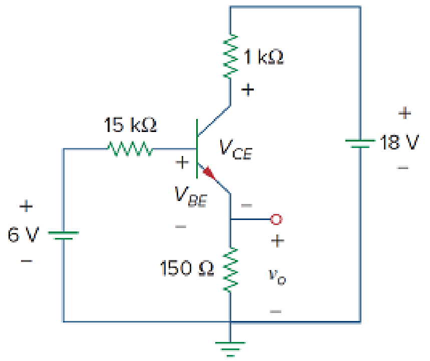
Chapter 3, Problem 12PP

Problem

For the transistor circuit in Fig. 3.42, let β = 100 and V_{BE} = 0.7 V. Determine v_o and V_{CE}.

Figure 3.42:



Step-by-step solution

Step 1 of 6

Refer to Figure 3.42 in the textbook.

Draw the circuit diagram with current directions and voltages as shown in Figure 1.

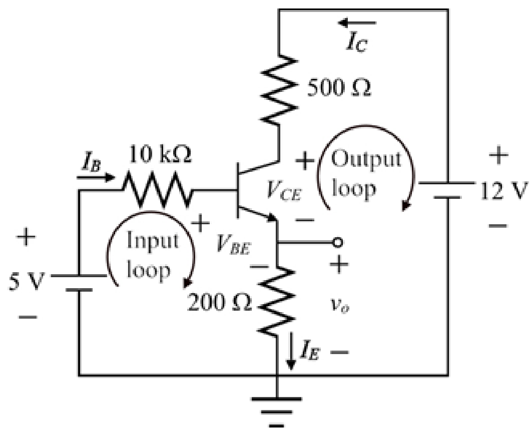


Figure 1

Step 2 of 6

The circuit shown in Figure 1, contains an *npn* transistor.

For the *npn* transistor,

$I_E = I_B + I_C$ ...... (1)

Write the expression for collector current I_C.

$$I_C = \beta I_B$$

Substitute 100 for β.

$$I_C = 100 I_B$$

Substitute $100 I_B$ for I_C in equation (1).

$$I_E = I_B + 100 I_B$$
$$I_E = 101 I_B$$

Step 3 of 6

Apply Kirchhoff's voltage law to the input loop.

$$-5 + \left(10 \times 10^3\right) I_B + V_{BE} + (200\ \Omega) I_E = 0$$

Substitute 0.7 V for V_{BE} and $101 I_B$ for I_E.

$$-5 + \left(10 \times 10^3\right) I_B + 0.7 + (200\ \Omega)(101 I_B) = 0$$
$$I_B \left(10 \times 10^3 + (200)(101)\right) = 4.3$$
$$30200 I_B = 4.3$$
$$I_B = 0.1424\ \text{mA}$$

Step 4 of 6

Write the expression for collector current I_C.

$$I_C = \beta I_B$$

Substitute 100 for β and $0.1424\ \text{mA}$ for I_B.

$$I_C = (100)(0.1424\ \text{mA})$$
$$= 14.24\ \text{mA}$$

Write the expression for emitter current I_E.

$$I_E = 101 I_B$$

Substitute $0.1424\ \text{mA}$ for I_B,

$$I_E = 101\left(0.1423 \times 10^{-3}\right)$$
$$= 14.38\ \text{mA}$$

Step 5 of 6

v_o is the voltage across $200\ \Omega$ resistance.

Write the expression for output voltage v_o using ohms law.

$$v_o = 200 I_E$$

Substitute $14.38\ \text{mA}$ for I_E.

$$\begin{aligned} v_o &= (200)\left(14.38 \times 10^{-3}\right) \\ &= 2.876\ \text{V} \end{aligned}$$

Therefore, the output voltage, v_o is $\boxed{2.876\ \text{V}}$.

Step 6 of 6

Apply Kirchhoff's voltage law to the outer loop.

$$-12 + 500 I_C + V_{CE} + v_o = 0$$

Substitute $2.876\ \text{V}$ for v_o and $14.24\ \text{mA}$ for I_C.

$$\begin{aligned} -12 + (500)\left(14.24 \times 10^{-3}\right) + V_{CE} + 2.876 &= 0 \\ V_{CE} &= 12 - 7.12 - 2.876 \\ V_{CE} &= 2.004\ \text{V} \end{aligned}$$

Therefore, the voltage, V_{CE} is $\boxed{2.004\ \text{V}}$.